INFRAPULSE

Rural Infrastructure Reliability Intelligence Platform

SDG 9 — Industry, Innovation and Infrastructure

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Build-ready specification for an AI-assisted development workflow

Core proposition: Infrastructure availability does not guarantee reliable service. InfraPulse combines availability, service-performance evidence, accessibility, and population impact to help identify reliability gaps and compare interventions.

Version 1.0 | Prepared 9 October 2026 | Hackathon MVP with production-minded extension path

1. Document purpose and status

This PRD defines the product scope, users, requirements, workflows, data handling, measurable acceptance criteria, and MVP boundaries for InfraPulse. It is intended to be given to an AI coding assistant or development team as the source of truth. Requirements labelled MVP are required for the first complete working prototype. Future features are explicitly separated so they do not silently expand the 24-hour build.

Evidence discipline: InfraPulse is a proposed decision-support layer, not an official government service. Scores and recommendations must be explainable. Any demo data that is simulated must be visibly labelled as simulated. Do not claim that a data feed is live unless the integration is implemented and tested.

2. Product vision

Vision: move rural infrastructure planning from “is an asset recorded?” toward “is the service usable and reliable, who is underserved, and which intervention should be considered first?”

InfraPulse aggregates geospatial baseline data and performance indicators across selected infrastructure dimensions. It calculates a configurable village reliability score, detects availability-versus-performance gaps, ranks villages for review, explains score drivers, and lets planners test hypothetical interventions. It complements source systems; it does not replace them.

3. Problem statement

Rural infrastructure records can establish that roads, broadband, electricity connections, public digital service points, schools or health facilities exist, but presence alone does not establish quality, continuity, seasonal accessibility, or practical access for residents. Information is distributed across sector-specific datasets, measured at different times, and may have different spatial units. Decision-makers can therefore struggle to compare service reliability across sectors, identify where recorded availability masks poor outcomes, estimate who is affected, and prioritize limited maintenance or investment resources.

4. Goals and non-goals

Goals	Non-goals for MVP
Create a map of demo villages and infrastructure dimensions.	Replace official systems such as Gram Manchitra, Mission Antyodaya, BharatNet or utility systems.
Calculate transparent scores from documented inputs.	Claim official government ratings or certified infrastructure assessments.
Expose missing, stale, low-confidence and simulated data.	Present simulated values as observed or government-verified data.
Rank villages with a visible reason breakdown.	Automatically approve, allocate or disburse public funds.
Compare user-defined intervention scenarios.	Claim predictive accuracy without validation data.

5. Target users and jobs-to-be-done

User	Primary job	Product value
District planner / administrator	Find villages needing investigation and compare interventions.	Ranked map, reasons, exportable evidence.
Infrastructure department analyst	Review sector-level performance and gaps.	Component scores, data provenance, filters.
Rural development / NGO program team	Identify underserved communities and track improvement.	Population-weighted priority and before/after scenarios.
Researcher / evaluator	Understand methodology and reproduce results.	Visible formula, source, date, confidence and assumptions.
Field verifier	Confirm whether reported conditions match reality.	Structured observation form and evidence status.

6. Product principles

• Explainability first: show which input values and weights produced each score.

• Data provenance: every record has source, collection date, spatial level and confidence.

• Honest uncertainty: missing data must not silently become zero or a favourable score.

• Decision support, not automated authority: recommendations require human review.

• Accessible design: keyboard-friendly controls, readable contrast, responsive layouts and plain-language explanations.

• Geographic caution: village boundaries, point locations and travel distances can be approximate; label them accordingly.

7. Scope and modules

Module	MVP behavior	Acceptance outcome
Overview dashboard	Show village count, score distribution, critical/high-priority count, data-quality notices.	All headline metrics match filtered dataset.
Interactive GIS map	Display villages with score-based styling; select village to open profile; filter by dimension/status.	Map and list use the same filtered records.
Village profile	Show population, component scores, score, reliability gaps, evidence dates, confidence and priority reasons.	Profile values trace back to the selected record.
Scoring engine	Normalize inputs and calculate a configurable composite score.	Same inputs/config produce same result; tests cover edge cases.
Gap detection	Compare availability indicator to reliability/performance indicator where both exist.	Gap is shown in points with direction and missing-data state.
Priority ranking	Rank based on need, people affected, criticality and alternatives, with visible factors.	Rank order is reproducible and explained.
What-if simulator	Let user alter one or more dimension scores or intervention assumptions and compare baseline vs scenario.	Baseline is unchanged; scenario result is recalculated transparently.
Data administration	Import CSV/GeoJSON, validate fields, show rejected rows and provenance.	Invalid rows produce actionable errors; no silent discard.
Reports/export	Download filtered results as CSV and a printable village summary.	Export matches active filters and current scores.
Feedback/evidence capture	Optional field note with date, dimension, text, optional photo URL and verification status.	Unverified reports are visibly distinguished from verified data.

8. Infrastructure dimensions and indicators

Dimension	Candidate indicators	Direction / notes
Digital connectivity	Availability; download/upload speed; uptime; latency; packet loss; public Wi-Fi/CSC access.	Higher uptime/speed generally better; normalize against documented target or percentile. Network tests are snapshots unless repeated.
Electricity	Connection availability; outage count; outage duration; critical facility backup.	Fewer/shorter outages are better. Use a stated observation period.
Road accessibility	Road condition; all-season passability; seasonal closure days; travel time to essential services; alternative route.	Shorter travel time and fewer closures are better. Do not treat distance alone as proof of road quality.
Public digital infrastructure	CSC/service-point presence; operational availability; public Wi-Fi usability; banking access.	Presence and operational status are distinct indicators.
Essential-service accessibility	Travel time/distance to healthcare, school, market, government and emergency services.	Prefer network travel time where reliable network data exists; otherwise label straight-line distance approximation.

9. Functional requirements

• FR-01: The application shall load without authentication in the demo environment, unless a deployment requires protected admin actions.

• FR-02: The user shall be able to filter villages by district/region, reliability category, infrastructure dimension and data-confidence status.

• FR-03: Selecting a map feature or table row shall open the same village profile.

• FR-04: Every score shall show component scores, weights, missing inputs and formula version.

• FR-05: The application shall distinguish observed, official-baseline, user-submitted, inferred and simulated values.

• FR-06: The gap module shall show availability and performance side by side; if either is missing, display “insufficient evidence” instead of inventing a gap.

• FR-07: The ranking module shall display a factor-level explanation for every priority score.

• FR-08: The simulator shall preserve the original baseline and show the assumptions changed by the user.

• FR-09: Data imports shall validate schema, coordinate ranges, duplicate IDs, numeric ranges and required fields.

• FR-10: Exports shall include record IDs, scores, component values, source labels, dates and methodology version.

• FR-11: The interface shall provide a data dictionary, methodology page and limitations notice.

• FR-12: All examples using simulated values shall carry a persistent “DEMO DATA — NOT OFFICIAL MEASUREMENTS” label.

10. Scoring specification

Use a transparent, configurable weighted score for the MVP; do not add machine learning unless labelled inputs and evaluation data exist. Suggested starting weights: digital connectivity 25%, electricity 20%, road accessibility 20%, public digital infrastructure 15%, essential-service accessibility 20%. These are product-design assumptions, not official weights. Keep them in a versioned configuration file.

Component normalization: transform each raw indicator to a 0–100 component score using a documented method. For “higher is better” indicators, a min-max transform can be used only when the reference bounds are meaningful and fixed. For “lower is better” indicators, invert the normalized result. Prefer policy/engineering thresholds or documented targets over min-max across a tiny demo dataset.

Composite score: overall = sum(weight_i × component_score_i) / sum(weights of valid components). If valid weights cover less than a configurable minimum coverage threshold (suggested 60%), show “insufficient data” instead of a final score. Also show coverage percentage. Never treat missing values as zero by default.

Availability–reliability gap: gap = availability_score − reliability_score, where both are on the same 0–100 scale and semantically comparable. Positive values indicate recorded availability exceeds observed reliability. Do not calculate this across incompatible units or different time periods without a caveat.

Prototype categories: 80–100 Highly reliable; 60–79 Adequate; 40–59 Vulnerable; 0–39 Critical. These thresholds are configurable and must be described as prototype categories, not official standards.

11. Priority score and intervention logic

Priority is not identical to low reliability. The MVP should combine (a) infrastructure deficit, (b) estimated people affected, (c) service criticality, (d) lack of alternatives, and (e) evidence confidence. Normalize each factor to 0–100 and define configurable weights, for example deficit 35%, people affected 25%, criticality 20%, lack of alternatives 10%, evidence confidence 10%. Because confidence is not need, display it separately and consider using it as a review/verification modifier rather than letting uncertain data alone inflate priority. Document this choice.

Intervention simulator must be scenario analysis, not a claim of causal prediction. A user changes a dimension score or cost/impact assumption; the engine recalculates the composite and priority score. If cost is entered, calculate score improvement per cost unit. Label results “scenario estimate based on user assumptions.” Do not invent a construction cost or predicted real-world outcome.

12. Data requirements and provenance

Field group	Required fields / rules
Identity	village_id (stable unique string), village_name, state, district; optional block/mandal, local government code.
Geography	latitude/longitude or GeoJSON geometry; coordinate system documented; validate ranges.
Population	population value, reference year, source, spatial unit; never imply census-current if older.
Indicator	indicator_name, raw_value, unit, normalized_score, observation_period, method.
Provenance	source_name, source_url or dataset ID, collected_at, source_date, licence/terms, evidence_type.
Quality	confidence_score or category, missing_reason, verification_status, reviewer/date where applicable.
Scoring	component_score, overall_score, score_version, weights_version, coverage_percent.
Feedback	report_id, village_id, dimension, description, created_at, verification_status; minimise personal data.

13. Data strategy for the hackathon

• Use authoritative public datasets where available and permitted; record the exact source and date.

• Use Gram Manchitra and Mission Antyodaya as existing systems/context, not as data to scrape in violation of site terms.

• Use OpenStreetMap data subject to its licence and attribution requirements for geographic context; check data completeness before use.

• Use TRAI MySpeed as a possible reference for how network quality can be measured; do not assume a public API exists.

• Where village-level outage or uptime data cannot be obtained, use clearly labelled simulated indicators for the demo and design the ingestion interface for future verified data.

• Keep each data row tagged as official baseline, observed measurement, community report, inferred estimate or simulation.

• Store source URL, observation date, spatial resolution and caveat with every metric.

14. UX requirements

• Plain-language labels: “Works reliably”, “Needs verification”, “Critical gap”; avoid implying official status.

• Map legend must explain score categories and the fact that categories are prototype thresholds.

• Provide a table alternative to the map for accessibility and low-bandwidth use.

• Show a visible “Data coverage” and “Last updated” area.

• Show score drivers and missing evidence without requiring users to inspect raw JSON.

• Make the demo responsive for laptop presentation and tablet/mobile review.

• Use neutral, professional visual design; avoid decorative AI-generated claims or fake government logos.

15. Security, privacy and governance

• Do not collect names, phone numbers, precise household coordinates or other personal data unless strictly necessary and consented.

• Aggregate community reports to village/dimension level for public display; do not expose reporter identity.

• Validate uploaded files and limit file size/type; escape rendered user text to prevent script injection.

• Use server-side secrets in environment variables; never commit API keys.

• Apply role protection to data import and administrative edits if those features are enabled.

• Keep audit fields for imports, score-version changes and manual overrides.

• Provide correction/verification workflow for disputed data and label unverified reports.

16. Non-functional requirements

Area	Requirement
Performance	For demo dataset of up to 5,000 village records, initial dashboard should be usable within 3 seconds on typical broadband; map layers should avoid rendering every complex feature at once.
Reliability	Core dashboard, scoring and simulator must work without external AI API access.
Accessibility	Keyboard-accessible controls, visible focus, sufficient contrast, semantic labels and table alternative.
Maintainability	Separate data ingestion, scoring, priority, API and UI; add tests for formula logic.
Observability	Structured server logs for failed imports and API errors; no secrets or personal data in logs.
Portability	Local setup documented with one-command development startup where practical; deployment environment variables documented.

17. Success metrics

• 100% of displayed scores show methodology version and data coverage.

• 100% of demo/simulated records carry a visible simulated-data label.

• At least 95% of valid fixture rows import successfully; invalid rows report exact reasons.

• Filtering, selection, ranking and exports return consistent village IDs and scores.

• Scoring tests pass for normal, boundary, missing-data and invalid-data cases.

• Users can identify the highest-priority village and explain the top three contributing factors within a two-minute demo.

• Intervention simulation produces a reproducible before/after score and clearly shows user assumptions.

18. Risks and mitigations

Risk	Mitigation
Existing systems already map and rank rural infrastructure.	Position InfraPulse as a complementary reliability evidence and scenario layer; cite overlap honestly.
Village-level reliability data unavailable.	Use labelled demo data; build provenance-first ingestion and field-verification workflow.
Different sources have different dates/spatial units.	Show dates/resolution and warn when comparing mismatched data.
Score gives false precision.	Publish formulas, weights, coverage and sensitivity; call scores indicative.
AI-generated explanation hallucinates.	Generate explanations from structured score drivers; require cited input fields; deterministic template first.
Judges question novelty.	Demonstrate the exact flow: gap → affected population → reasoned priority → scenario comparison.
Scope too large for 24 hours.	Prioritize one district/region, a small validated dataset, five dimensions, core score/map/ranking/simulator.

19. Acceptance checklist

• Fresh clone can be installed using documented steps.

• Seed/demo data loads and is labelled as simulated where applicable.

• Map, list and village detail remain consistent under filters.

• Missing data yields a coverage warning rather than an invented score.

• Gap calculation only runs when both comparable values exist.

• Priority explanation names actual contributing factors.

• Simulator changes only the scenario state, not the baseline record.

• CSV export matches visible filtered results.

• All key scoring and validation tests pass.

• Methodology, source links, limitations and demo script are included in the README.

20. Sources and research notes

These sources establish the existing landscape and measurement context. They do not prove that every source provides an open API or village-level reliability data. Verify current terms and access before implementation.

Gram Manchitra — official Gram Panchayat GIS platform

Mission Antyodaya — village/GP survey, rankings and gap reports

BharatNet — rural broadband infrastructure programme

TRAI Apps / MySpeed — network performance measurements

World Bank Rural Access Index — SDG 9.1.1

SDG indicator 9.1.1 methodology and limitations

Research-derived positioning: Mission Antyodaya already collects indicators across 21 development sectors and generates GP/village rankings and gap reports; Gram Manchitra already provides a GIS, asset, proximity and gap-report interface; TRAI MySpeed measures network quality metrics; the World Bank RAI measures rural access to all-season roads. InfraPulse must therefore not claim that rural infrastructure mapping, gap analysis, scoring or connectivity measurement are individually novel. Its proposed differentiation is the product workflow that joins multi-dimensional reliability evidence, data confidence, population/service impact, transparent ranking and user-controlled intervention scenarios in one traceable interface. Validate the actual overlap by testing each cited system and reviewing its current features before making competition claims.